

ASU Macro Qual Module 3

Asset Pricing Theory

PhD Qual Lecture Notes Builder Packet

Built for ASU Macro Theory Qual Preparation

Course focus	ECN 713 / ASU Macro Theory Qual
Module	3: Asset Pricing Theory
Target weaknesses	formulation and setup; economic intuition; variations; dynamics and comparative statics
Study use	two 3-hour blocks plus one 90-minute closed-form drill block
Core output	source inventory, blueprint, lecture packet, exercises, answer sketches, practice map

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1 Source inventory

Accessible source base and extraction notes

Source	Type	Topics used	Exam relevance / extraction notes
MACRO_MANIFEST.md	Project manifest	Module 3 coverage requirements; source hierarchy; notation conventions; PS2 anchors; required derivations.	Governs the packet. It requires Lucas trees, fixed asset supply, SDF, Markov taxes, CRRA growth, war risk, bonds, expected returns, equity premia, exercises, traps, and past-qual mapping.
SKILL.md	PhD qual note-building skill	Standard packet structure; asset-pricing derivation checklist; exercise and answer-sketch requirements; TeX style.	Governs output format: source inventory, blueprint, canonical derivations, closed-form catalog, exam variations, traps, exercise bank, answer sketches, and final checklist.
GitHub folder: https://github.com/Bonorinoa/Quals-KB/tree/main/Macroeconomics/Asset%20Pricing	Repository source folder	Confirms the Asset Pricing folder contains Natalia's asset-pricing notes and PS2.	Used to verify folder-level source inventory. Local uploaded copies were used for detailed extraction.
2_Asset Pricing.pdf	Natalia lecture notes	Lucas-tree environment; sequence and recursive consumer problem; fixed tree supply; FOC and envelope; stochastic Euler equation; SDF; risk-free rate; covariance risk premium; linear utility; independent and correlated shocks; equity premium puzzle.	Professor framing and notation source. Parsed text was available; no diagrams were needed. Some PDF line breaks were cleaned manually in formulas.

Source	Type	Topics used	Exam relevance / extraction notes
PS2-6.pdf	Current 2026 problem set	Markov dividend and tax states; dividend taxes with rebates; log two-tax-state functional equations; persistence comparison; deterministic dividend growth under CRRA; expected return vs $1/\beta$; war-risk pricing and risk-aversion comparison.	Current-course anchor. This is the highest priority source after past exams.
2016-06...tex	Past ASU Macro qual	Two-tree Lucas economy; recursive problem; RCE; functional equations; log iid dividend with one constant tree; prices higher in high-dividend states.	Relevant past-qual asset-pricing problem. Local TeX was readable.
2017-06...tex	Past ASU Macro qual	Two negatively correlated trees; constant aggregate dividends; RCE; explicit prices; price-sum result; binary persistence comparative statics.	High-yield because it forces the student to separate aggregate consumption from individual dividends. Local TeX was readable.
2018-06...tex	Past ASU Macro qual	Time-varying net supply of a dividend asset; external investor demand; odd/even aggregate state; linear versus strictly concave utility price comparison.	High-yield setup trap: equilibrium holdings are time-varying, so aggregate state and market clearing must be imposed carefully. Local TeX was readable.
2019-06...tex	Past ASU Macro qual	Two countries; each country trades only domestic stock; common one-period bond; recursive problems; stationary RCE; FOC, envelope, Euler; alternating dividends closed form.	Useful bond-and-segmentation variation. Local TeX was readable.
2020-06...tex	Past ASU Macro qual	Search model with asset market and one-period bond.	Not used as an asset-pricing anchor; better assigned to Search/RCE modules.

Source	Type	Topics used	Exam relevance / extraction notes
2021-06...tex	Past ASU Macro qual	Lucas tree with tax on dividends plus beginning-of-period asset value; stationary price function; recursive proof prices decrease in tax; log closed form; efficiency claim.	High-yield because it tests tax timing and the distinction between price distortions and allocation inefficiency. Local TeX was readable.
2022-06...tex	Past ASU Macro qual	Project-choice risk and returns, not Lucas-tree asset pricing.	Not directly used except as a general dynamic-risk comparison.
2023-06...tex	Past ASU Macro qual	Lucas tree with risky stock and risk-free bond; preferences over consumption plus bond holdings; log closed forms; expected stock return, risk-free rate, equity premium and liquidity value κ .	High-yield recent asset-pricing problem and included because the manifest mentions bond-in-utility/liquidity value when relevant.
2024-06...tex	Past ASU Macro qual	No direct asset-pricing problem.	Not used for Module 3.

Pattern analysis from the source base

The asset-pricing questions repeat a small number of commands. A strong answer almost always does the following:

- P1.** State timing: beginning-of-period asset holdings receive current dividends, then the agent trades ex-dividend assets at current prices.
- P2.** Write the individual recursive problem before imposing equilibrium.
- P3.** Derive the FOC with respect to next-period asset holdings and the envelope condition with respect to beginning-of-period asset holdings.
- P4.** Combine FOC and envelope, then impose fixed asset supply and feasibility to obtain an equilibrium pricing equation.
- P5.** Use the stochastic discount factor $M_{t+1} = \beta u'(c_{t+1})/u'(c_t)$ to interpret prices, risk premia, and risk-free rates.
- P6.** For log or CRRA cases, guess the price form, substitute, match coefficients, state convergence conditions, and interpret comparative statics.
- P7.** For variations, identify what changes in the payoff: dividend, tax, bond payoff, liquidity payoff, aggregate supply, or transition probabilities.

Qualifier trap

The most common point loss is substituting equilibrium too early. Write the household problem with individual holdings x and choices y first. Only after the FOC and envelope should you set $x = y = 1$ or $y = \bar{y}(s)$.

2 Detailed blueprint

2.1 Module objective

By the end of this module, the student should be able to take any Lucas-tree-style ASU macro qualifier question, write the timing-correct recursive household problem, derive the FOC-envelope-Euler system, impose equilibrium market clearing, solve log/CRRA closed forms, and defend comparative statics formally and intuitively.

2.2 Prerequisites

- Basic dynamic programming: state, control, Bellman equation, FOC, envelope.
- Markov chains and conditional expectations.
- Consumer theory: marginal utility, strict concavity, CRRA/log utility.
- Equilibrium definitions: individual optimization, market clearing, consistency.
- Algebra for two-state linear systems and guess-and-verify solutions.

2.3 Canonical models and where they come from

Canonical model	Core setup	Main deliverable	Main source anchor
Generic k -tree Lucas economy	k fixed-supply trees, Markov dividends, representative household	Bellman, FOC, envelope, Euler, SDF, price operator	Natalia notes; 2016; 2017
One-tree log Lucas economy	One fixed-supply tree, $c = z$, $u(c) = \log c$	$p(z) = \frac{\beta}{1-\beta}z$ and expected return formula	Natalia notes; PS2
Two-tree constant aggregate consumption	Dividends move across trees but aggregate dividends are constant	Prices equal discounted expected dividends; $p_1 + p_2$ constant; persistence spread formula	2017 qual
Log iid two-tree with one constant tree	Tree 1 stochastic iid, tree 2 constant, $c = x + 1$	Multiplicative current-consumption price formulas	2016 qual
Dividend-tax Lucas tree	Markov dividends and Markov dividend taxes with rebates	Tax-state pricing equations and persistence comparison	PS2

Canonical model	Core setup	Main deliverable	Main source anchor
Tax on dividends plus asset value	Current tax on beginning-of-period taxable income	Price operator with factor $1 - \tau$; log closed form decreasing in tax	2021 qual
Deterministic dividend growth	$d_{t+1} = \gamma d_t$, CRRA utility	Price-dividend ratio, growth comparative statics, return vs $1/\beta$	PS2
War/disaster risk	One-time permanent fall from z to sz	Before/after price, expected return, risk-aversion comparison	PS2
Bond and bond-in-utility	Risky stock plus one-period risk-free bond; optional liquidity value κ	Bond Euler equation, risk-free rate, equity premium effect of liquidity	2019; 2023
Time-varying net supply	External investors change household net supply over odd/even periods	Aggregate-state setup and odd/even price ranking under concavity	2018 qual

2.4 Closed-form catalog

The packet derives and reuses these formulas.

C1. Generic SDF price: $p_i(z) = \mathbb{E}[M(z, z')(d'_i + p_i(z')) \mid z]$.

C2. Log one-tree: $p(z) = \frac{\beta}{1-\beta} z$.

C3. Two-tree constant aggregate consumption: $p_1(z) + p_2(z) = \frac{\beta C}{1-\beta}$.

C4. Binary persistence spread under constant marginal utility: $\Delta p = \frac{\beta \rho}{1-\beta \rho} \Delta d$, where $\rho = 2\pi - 1$.

C5. Markov tax-state spread in PS2: $p_L - p_H = \frac{\beta(2\theta-1)\bar{\tau}\bar{z}}{1-\beta(2\theta-1)}$.

C6. CRRA growing dividend: $p_t = \nu d_t$, $\nu = \frac{\beta \gamma^{1-\sigma}}{1-\beta \gamma^{1-\sigma}}$.

C7. Deterministic growth return: $R = \gamma^\sigma / \beta$.

C8. War-risk CRRA before-war price ratio: $p_B/z = \frac{\beta[(1-\pi)+\pi s^{1-\sigma}/(1-\beta)]}{1-\beta(1-\pi)}$.

C9. Standard risk-free bond: $q(z) = \mathbb{E}[M(z, z') \mid z]$.

C10. Bond-in-utility log price: $q(z) = \beta(1 + \kappa)z\mathbb{E}[1/z' \mid z]$.

2.5 Exercise plan

The exercise bank is organized into four categories: mechanical setup drills, closed-form/proof drills, past-qual applications, and trap/economist-claim questions. Answer sketches are included, not just final formulas, because the target weakness is setup and intuition.

3 Module map

Subtopic	Why it matters	Exam/practice appearances	Core technique	Main trap
Ex-dividend timing	Determines the correct budget constraint and payoff in the Euler equation.	All Lucas-tree problems.	Current holdings receive current dividends; purchases today receive next dividends.	Putting current dividends on the payoff to today's buyer.
Recursive household problem	Quads ask this directly before any pricing algebra.	PS2; 2016; 2017; 2018; 2019; 2021; 2023.	State x plus aggregate state; choice y ; price function taken as given.	Imposing $x = 1$ before deriving FOC/envelope.
FOC-envelope-Euler	The core derivation.	Every asset-pricing source.	FOC in y_i , envelope in x_i , evaluate at equilibrium.	Forgetting the envelope includes dividend plus resale price.
SDF and risk premia	Converts Euler equations into intuition.	Natalia notes; 2023.	$M = \beta u'(c')/u'(c)$, $1 = \mathbb{E}[MR]$.	Saying risk lowers price without checking covariance with M .

Subtopic	Why it matters	Exam/practice appearances	Core technique	Main trap
Log one-tree	Fast closed form used repeatedly.	Natalia notes; PS2; 2021; 2023.	Let $f = p/z$; solve $f = \beta + \beta \mathbb{E}f$.	Treating Markov persistence as relevant when log one-tree makes f constant.
Two-tree economies	Tests aggregate versus individual dividend risk.	2016; 2017.	Use aggregate consumption in u' , not individual tree dividends.	Pricing each tree as if it were the only consumption source.
Taxes and rebates	Tests timing and equilibrium resource accounting.	PS2; 2021.	Include after-tax payoff and rebate; in equilibrium $c = z$ if tax is rebated.	Using current tax instead of next tax in the payoff.
CRRA growth	Tests normalized prices and comparative statics.	PS2.	Guess $p_t = \nu d_t$; solve for ν .	Saying higher growth always raises price.
War/disaster risk	Tests state-contingent marginal utility.	PS2.	Separate before-war and after-war states; compare to no-war economy by σ .	Confusing expected return with required return.

Subtopic	Why it matters	Exam/practice appearances	Core technique	Main trap
Risk-free bonds	Tests bond Euler and equity premium.	2019; 2023.	Add qb' to the budget and derive bond FOC.	Missing the direct utility effect of bond holdings.
Time-varying supply	Tests aggregate state and market clearing.	2018.	Track beginning holdings and next holdings separately.	Assuming fixed supply equals one in every state.

4 Learning objectives

After studying this packet, you should be able to:

- LO1.** Given any fixed-supply Lucas tree problem, write the household Bellman equation with the correct individual and aggregate state variables.
- LO2.** Derive the FOC, envelope condition, and stochastic Euler equation without skipping the individual optimization step.
- LO3.** Define a stationary recursive competitive equilibrium for one-tree, multi-tree, bond, tax, and time-varying-supply economies.
- LO4.** Convert Euler equations into SDF, risk-free rate, expected return, and risk premium formulas.
- LO5.** Solve log and CRRA closed forms by guess-and-verify.
- LO6.** Rank prices across tax states, dividend states, persistence parameters, growth rates, war risk, and liquidity parameters.
- LO7.** Explain comparative statics in terms of income effects, substitution effects, state-contingent marginal utility, and equilibrium market clearing.

5 Notation and timing conventions

5.1 Default notation

- Time is discrete, $t = 0, 1, 2, \dots$
- Discount factor: $\beta \in (0, 1)$.
- Exogenous Markov state: $z \in Z$; next state is z' .
- Asset price is ex-dividend: $p(z)$ or $p_i(z)$.
- A one-period bond price is $q(z)$ and pays one unit next period.
- Beginning-of-period asset holdings are x ; next-period holdings chosen today are y .
- Equilibrium fixed supply is usually one share, so $x = y = 1$ after imposing market clearing.
- Stochastic discount factor:

$$M(z, z') = \beta \frac{u'(c')}{u'(c)}.$$

5.2 Timing that should appear on the exam paper

At the beginning of period t , state z_t is observed. A household enters with shares x_t that were purchased last period. These shares pay current dividends $d(z_t)x_t$. Then the household trades shares at the ex-dividend price $p(z_t)$ and leaves the period holding $x_{t+1} = y_t$.

For one asset, the budget constraint is

$$c + p(z)y \leq [d(z) + p(z)]x.$$

For k trees, write vectors:

$$c + p(z) \cdot y \leq [d(z) + p(z)] \cdot x.$$

Qualifier trap

If the problem says the asset is traded after dividends and taxes are realized, the asset purchased today does not receive today's dividend. Its payoff in the Euler equation is next period's after-tax dividend plus next period's ex-dividend resale price.

6 Exam workflow: from words to pricing equation

- Step 1. List states.** Individual state: holdings entering the period, such as x and possibly bond holdings b . Aggregate state: exogenous Markov variables and any aggregate endogenous variable needed for prices, such as dividend state, tax state, parity, or beginning aggregate supply.
- Step 2. Write the budget.** Use ex-dividend timing. Include taxes, rebates, bond payments, and supply changes before simplifying.
- Step 3. Write the Bellman equation.** The household takes price functions as given.
- Step 4. FOC in next holdings.** Differentiate with respect to y or b' .
- Step 5. Envelope in current holdings.** Differentiate with respect to x or b .
- Step 6. Combine.** Substitute the envelope one period ahead into the FOC.
- Step 7. Impose equilibrium.** Fixed asset supply, bond zero net supply, feasibility, and government budget.
- Step 8. Solve or compare.** Use log/CRRA guesses, two-state linear systems, monotone operators, or SDF covariance intuition.

7 Canonical model 1: generic Lucas-tree economy

7.1 Environment

There is one nonstorable consumption good and k productive units, called trees. Tree i yields dividend z_i in the current period. The dividend vector is $z = (z_1, \dots, z_k) \in Z \subset \mathbb{R}_+^k$ and follows a Markov transition $Q(z, dz')$. Each tree has one perfectly divisible share in fixed supply. A share of tree i is a claim to all future dividends of tree i .

The representative household has expected utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where u is increasing, strictly concave, and differentiable.

7.2 Recursive household problem

Let $x = (x_1, \dots, x_k)$ be the household's beginning-of-period shares and let y be next-period shares. For a candidate stationary price function $p : Z \rightarrow \mathbb{R}_+^k$, define consumption as

$$c = z \cdot x + p(z) \cdot (x - y).$$

The recursive problem is

$$V(x, z) = \max_{y \in \Gamma(x, z)} \left\{ u(z \cdot x + p(z) \cdot (x - y)) + \beta \int V(y, z') Q(z, dz') \right\},$$

where

$$\Gamma(x, z) = \{y \geq 0 : p(z) \cdot y \leq [z + p(z)] \cdot x\}.$$

7.3 FOC and envelope

Assume an interior optimum. The FOC for y_i is

$$u'(c) p_i(z) = \beta \int V_{x_i}(y, z') Q(z, dz').$$

The envelope condition for x_i is

$$V_{x_i}(x, z) = u'(c)[z_i + p_i(z)].$$

Exam tip

The FOC has current price $p_i(z)$ on the left. The envelope has dividend plus resale price, $z_i + p_i(z)$, because beginning holdings receive current dividends and can be sold at the current ex-dividend price.

7.4 Equilibrium pricing equation

In equilibrium, all shares are held and all output is consumed:

$$x = y = \mathbf{1}, \quad c(z) = \sum_{j=1}^k z_j.$$

Substituting the envelope condition into the FOC gives the stochastic Euler equation

$$u'(c(z))p_i(z) = \beta \int u'(c(z'))[z'_i + p_i(z')]Q(z, dz'), \quad i = 1, \dots, k.$$

Equivalently,

Reusable formula

$$p_i(z) = \mathbb{E} [M(z, z')\{z'_i + p_i(z')\} \mid z], \quad M(z, z') = \beta \frac{u'(c(z'))}{u'(c(z))}.$$

Economic intuition

The asset is valuable when it pays in states with high marginal utility. If a payoff arrives in a high-consumption state, $u'(c')$ is low, so that payoff is discounted more heavily. This is the source of risk premia in a representative-agent Lucas tree economy.

7.5 Price operator and uniqueness

Define

$$f_i(z) = u'(c(z))p_i(z), \quad h_i(z) = \beta \int u'(c(z'))z'_i Q(z, dz').$$

Then the Euler equation becomes

$$f_i(z) = h_i(z) + \beta \int f_i(z')Q(z, dz').$$

The operator

$$(T_i f_i)(z) = h_i(z) + \beta \int f_i(z')Q(z, dz')$$

is a contraction under the sup norm when restricted to bounded continuous functions. Hence there is a unique bounded continuous f_i and therefore a unique price function

$$p_i(z) = \frac{f_i(z)}{u'(c(z))}.$$

7.6 Expected returns and risk-free rate

The gross return on tree i bought in state z and sold next period is

$$R_i(z, z') = \frac{z'_i + p_i(z')}{p_i(z)}.$$

The Euler equation is

$$1 = \mathbb{E}[M(z, z')R_i(z, z') \mid z].$$

A one-period risk-free bond that costs $q(z)$ and pays one unit next period satisfies

$$q(z) = \mathbb{E}[M(z, z') \mid z], \quad R^f(z) = \frac{1}{q(z)}.$$

Using $1 = \mathbb{E}[MR]$ and $R^f = 1/\mathbb{E}[M]$,

Reusable formula

$$\mathbb{E}[R_i \mid z] - R^f(z) = -R^f(z) \operatorname{Cov}(M, R_i \mid z).$$

Economic intuition

If an asset's return is high exactly when consumption is high, then it has negative covariance with M because marginal utility is low in high-consumption states. The asset is a bad hedge, so its expected return must be high. If it pays in bad times, it is insurance and can have a low expected return.

8 Canonical model 2: log one-tree Lucas economy

8.1 Setup

Let there be one tree in fixed supply. Its dividend is $z > 0$ and follows a Markov process. In equilibrium, $c = z$. With $u(c) = \log c$, marginal utility is $u'(c) = 1/c = 1/z$.

The pricing equation is

$$\frac{p(z)}{z} = \beta \mathbb{E} \left[\frac{z' + p(z')}{z'} \mid z \right].$$

Define $f(z) = p(z)/z$. Then

$$\begin{aligned} f(z) &= \beta \mathbb{E}[1 + f(z') \mid z] \\ &= \beta + \beta \mathbb{E}[f(z') \mid z]. \end{aligned}$$

The unique bounded solution is constant:

$$f(z) = \frac{\beta}{1 - \beta}.$$

Therefore

Reusable formula

$$p(z) = \frac{\beta}{1 - \beta} z.$$

8.2 Expected stock return and risk-free rate

The stock return is

$$R(z, z') = \frac{z' + p(z')}{p(z)} = \frac{z'/(1 - \beta)}{\beta z/(1 - \beta)} = \frac{z'}{\beta z}.$$

Thus

$$\mathbb{E}[R \mid z] = \frac{\mathbb{E}[z' \mid z]}{\beta z}.$$

The risk-free bond price is

$$q(z) = \beta \mathbb{E} \left[\frac{z}{z'} \mid z \right], \quad R^f(z) = \frac{1}{\beta z \mathbb{E}[1/z' \mid z]}.$$

Economic intuition

The log one-tree price is proportional to current output. A high current dividend means high current consumption and low current marginal utility. Households are willing to pay more today to move wealth into the future. This income effect exactly delivers a constant price-dividend ratio $\beta/(1 - \beta)$.

Qualifier trap

In the log one-tree case, Markov persistence does not affect the price-dividend ratio. It affects expected returns through $\mathbb{E}[z' \mid z]/z$, not the formula $p(z)/z = \beta/(1 - \beta)$.

9 Linear utility benchmark

With linear utility, marginal utility is constant. The pricing equation becomes

$$p_i(z) = \beta \mathbb{E}[z'_i + p_i(z') \mid z].$$

Iterating forward and imposing boundedness gives

Reusable formula

$$p_i(z) = \sum_{s=1}^{\infty} \beta^s \mathbb{E}[z_{i,t+s} \mid z_t = z].$$

Economic intuition

Linear utility removes risk adjustment. Prices are expected discounted dividends. Concavity changes prices only through state-contingent marginal utility.

10 Canonical model 3: two-tree economies

10.1 General two-tree equations

Let the two trees have dividends $d_1(z)$ and $d_2(z)$. Aggregate consumption is

$$c(z) = d_1(z) + d_2(z).$$

The pricing equations are

$$p_i(z) = \mathbb{E} \left[\beta \frac{u'(c(z'))}{u'(c(z))} \{d_i(z') + p_i(z')\} \mid z \right], \quad i = 1, 2.$$

Qualifier trap

The marginal utility term uses aggregate consumption, not tree i 's dividend. This is the central two-tree setup trap.

10.2 Constant aggregate consumption: negatively correlated trees

Suppose

$$d_1(z) = z, \quad d_2(z) = 1 - z, \quad z \in (0, 1).$$

Then aggregate consumption is constant: $c(z) = 1$. Marginal utility is constant, so

$$p_1(z) = \beta \mathbb{E}[z' + p_1(z') \mid z], \quad p_2(z) = \beta \mathbb{E}[1 - z' + p_2(z') \mid z].$$

Adding the two equations,

$$p_1(z) + p_2(z) = \beta \mathbb{E}[1 + p_1(z') + p_2(z') \mid z].$$

The unique bounded solution has constant sum

Reusable formula

$$p_1(z) + p_2(z) = \frac{\beta}{1 - \beta}.$$

Economic intuition

The two trees only reshuffle dividends across assets. The aggregate endowment is constant, so total equity value is constant. A high-dividend state for tree 1 makes tree 1 more valuable and tree 2 less valuable, but the sum is fixed.

10.3 Binary persistence spread

Let $z \in \{H, L\}$, where $H = \bar{z} > 1/2$ and $L = 1 - \bar{z}$. Let

$$\Pr(H \mid H) = \Pr(L \mid L) = \pi > 1/2, \quad \rho = 2\pi - 1.$$

For tree 1, define $\Delta p = p_1(H) - p_1(L)$ and $\Delta d = H - L = 2\bar{z} - 1$. Subtracting the two pricing equations gives

$$\Delta p = \beta \rho (\Delta d + \Delta p),$$

so

Reusable formula

$$\Delta p = \frac{\beta \rho}{1 - \beta \rho} \Delta d.$$

An increase in persistence π raises ρ and therefore increases the price spread. Tree 1's price responds more to the current dividend state because the current state is more informative about future dividends.

10.4 Log iid tree plus constant tree

This is the 2016-style variant. Tree 1 has iid dividend $x \in \{x^H, x^L\}$ and tree 2 has constant dividend 1. Aggregate consumption is $c = x + 1$. With log utility,

$$\frac{p_i(x)}{x+1} = \beta \mathbb{E} \left[\frac{d_i(x') + p_i(x')}{x' + 1} \right].$$

Because x' is iid, the right side does not depend on current x . Let

$$A_1 = \mathbb{E} \left[\frac{x'}{x' + 1} \right], \quad A_2 = \mathbb{E} \left[\frac{1}{x' + 1} \right].$$

Guess $p_i(x) = a_i(x + 1)$. Then

$$a_1 = \beta A_1 + \beta a_1,$$

$$a_2 = \beta A_2 + \beta a_2.$$

Hence

Reusable formula

$$p_1(x) = \frac{\beta}{1 - \beta} (x + 1) \mathbb{E} \left[\frac{x'}{x' + 1} \right], \quad p_2(x) = \frac{\beta}{1 - \beta} (x + 1) \mathbb{E} \left[\frac{1}{x' + 1} \right].$$

Both prices are higher when current x is higher because current aggregate consumption $x + 1$ is higher, current marginal utility is lower, and households bid up assets to move resources into future states.

11 Canonical model 4: taxes and rebates

11.1 Dividend tax with Markov tax state

Let the state be $s = (z, \tau)$, where dividends follow $P(z, z')$ and tax rates follow $Q(\tau, \tau')$. The government taxes dividends at rate τ and rebates tax revenue lump-sum. A household with beginning shares x chooses next shares y :

$$c + p(s)y \leq [p(s) + (1 - \tau)z]x + T(s).$$

The Bellman equation is

$$V(x, s) = \max_y \{u(c) + \beta \mathbb{E}[V(y, s') \mid s]\}$$

subject to the budget above. The FOC is

$$u'(c)p(s) = \beta \mathbb{E}[V_x(y, s') \mid s],$$

and the envelope condition is

$$V_x(x, s) = u'(c)[p(s) + (1 - \tau)z].$$

In equilibrium, $x = y = 1$, $T(s) = \tau z$, and $c = z$. Therefore

Reusable formula

$$u'(z)p(z, \tau) = \beta \sum_{z', \tau'} P(z, z')Q(\tau, \tau')u'(z')[(1 - \tau')z' + p(z', \tau')].$$

Exam tip

The after-tax dividend in the payoff is $(1 - \tau')z'$, not $(1 - \tau)z$. The asset bought today pays next period's after-tax dividend.

11.2 Two tax states, constant dividends, log utility

Let $z = \bar{z}$ be constant, $u(c) = \log c$, $\tau_L = 0$, $\tau_H = \bar{\tau} \in (0, 1)$, and

$$Q(\tau_L, \tau_L) = Q(\tau_H, \tau_H) = \theta.$$

Because consumption is constant, marginal utility cancels. Let $p_L = p(\tau_L)$ and $p_H = p(\tau_H)$. The pricing equations are

$$\begin{aligned} p_L &= \beta [\theta(\bar{z} + p_L) + (1 - \theta)((1 - \bar{\tau})\bar{z} + p_H)], \\ p_H &= \beta [(1 - \theta)(\bar{z} + p_L) + \theta((1 - \bar{\tau})\bar{z} + p_H)]. \end{aligned}$$

Subtracting gives

$$p_L - p_H = \beta(2\theta - 1)[\bar{\tau}\bar{z} + p_L - p_H],$$

so

Reusable formula

$$p_L - p_H = \frac{\beta(2\theta - 1)\bar{\tau}\bar{z}}{1 - \beta(2\theta - 1)}.$$

Economic intuition

The zero-tax state has a higher price only if the tax state is persistent enough, $\theta > 1/2$. If $\theta < 1/2$, being in the zero-tax state predicts a high chance of moving to the high-tax state next period, so the current zero-tax state can have a lower price. The price is about future after-tax payoffs, not the current tax label.

11.3 Tax on dividends plus beginning asset value

A 2021-style variant taxes income defined as dividends plus the value of assets owned at the beginning of the period. With tax rate τ and rebate $T(z)$, the household budget is

$$c + p(z)y \leq (1 - \tau)[z + p(z)]x + T(z).$$

The FOC is the same in form:

$$u'(c)p(z) = \beta \mathbb{E}[V_x(y, z') \mid z].$$

The envelope condition is different:

$$V_x(x, z) = u'(c)(1 - \tau)[z + p(z)].$$

In equilibrium, $T(z) = \tau[z + p(z)]$ and $c = z$. Therefore

Reusable formula

$$u'(z)p(z) = \beta(1 - \tau)\mathbb{E}[u'(z')(z' + p(z')) \mid z].$$

With log utility, guess $p(z) = az$. Then

$$a = \beta(1 - \tau)(1 + a),$$

so

Reusable formula

$$p(z) = \frac{\beta(1 - \tau)}{1 - \beta(1 - \tau)}z.$$

The coefficient is strictly decreasing in τ :

$$\frac{\partial}{\partial \tau} \left[\frac{\beta(1 - \tau)}{1 - \beta(1 - \tau)} \right] = -\frac{\beta}{[1 - \beta(1 - \tau)]^2} < 0.$$

Economic intuition

The tax lowers the private after-tax payoff to holding the tree, so the private asset price falls. But if the tree supply is fixed and tax revenue is rebated lump-sum, the aggregate allocation $c = z$ is unchanged. Lower prices do not by themselves imply allocation inefficiency.

12 Canonical model 5: deterministic dividend growth and CRRA pricing

12.1 Setup

There is one tree with deterministic dividend growth

$$d_{t+1} = \gamma d_t, \quad \gamma > 1.$$

The tree is in fixed supply one, so $c_t = d_t$. Utility is CRRA:

$$U(c) = \begin{cases} \frac{c^{1-\sigma}}{1-\sigma}, & \sigma > 0, \sigma \neq 1, \\ \log c, & \sigma = 1. \end{cases}$$

The Euler equation is

$$p_t = \beta \left(\frac{d_{t+1}}{d_t} \right)^{-\sigma} (d_{t+1} + p_{t+1}).$$

12.2 Closed-form price

Guess $p_t = \nu d_t$. Then

$$\begin{aligned} \nu d_t &= \beta \gamma^{-\sigma} (\gamma d_t + \nu \gamma d_t) \\ &= \beta \gamma^{1-\sigma} (1 + \nu) d_t. \end{aligned}$$

Let $A = \beta \gamma^{1-\sigma}$. If $A < 1$, then

Reusable formula

$$\nu = \frac{A}{1-A} = \frac{\beta \gamma^{1-\sigma}}{1 - \beta \gamma^{1-\sigma}}, \quad p_t = \frac{\beta \gamma^{1-\sigma}}{1 - \beta \gamma^{1-\sigma}} d_t.$$

The price grows at the same gross rate as dividends:

$$\frac{p_{t+1}}{p_t} = \gamma.$$

12.3 Growth comparative statics

For a fixed current dividend d_t , the sign of $\partial p_t / \partial \gamma$ is the sign of $1 - \sigma$:

$$\frac{\partial \nu}{\partial \gamma} = \frac{\beta(1-\sigma)\gamma^{-\sigma}}{[1 - \beta\gamma^{1-\sigma}]^2}.$$

Thus:

- if $\sigma < 1$, higher growth raises the price-dividend ratio;
- if $\sigma = 1$, higher growth does not change the price-dividend ratio;
- if $\sigma > 1$, higher growth lowers the price-dividend ratio.

Economic intuition

Higher growth raises future dividends, which tends to raise price. But higher growth also raises future consumption, which lowers future marginal utility and increases discounting. Risk aversion σ determines which force dominates.

12.4 Return

The gross return is deterministic:

$$R_{t+1} = \frac{d_{t+1} + p_{t+1}}{p_t} = \frac{\gamma d_t(1 + \nu)}{\nu d_t} = \gamma \frac{1 + \nu}{\nu}.$$

Since $\nu = A/(1 - A)$, $(1 + \nu)/\nu = 1/A$, so

Reusable formula

$$R_{t+1} = \frac{\gamma}{\beta \gamma^{1-\sigma}} = \frac{\gamma^\sigma}{\beta}.$$

For $\gamma > 1$ and $\sigma > 0$, $R_{t+1} > 1/\beta$. Also

$$\frac{\partial R_{t+1}}{\partial \gamma} = \frac{\sigma \gamma^{\sigma-1}}{\beta} > 0$$

when $\sigma > 0$.

Qualifier trap

Do not say “higher growth raises asset prices” without conditioning on σ and on whether current dividends are held fixed. The price level p_t grows over time with d_t , but the price-dividend ratio may fall with γ when $\sigma > 1$.

13 Canonical model 6: war and disaster risk

13.1 Setup

There is one tree. Before war, dividends are z . War arrives with probability $\pi \in [0, 1)$ each period, can occur only once, and permanently reduces dividends to sz , where $s \in (0, 1)$. Let p_A denote the after-war price and p_B the before-war price.

13.2 After-war price

After war has occurred, consumption is constant at sz . The Euler equation is

$$p_A = \beta(sz + p_A),$$

so

Reusable formula

$$p_A = \frac{\beta}{1 - \beta}sz, \quad R_A = \frac{sz + p_A}{p_A} = \frac{1}{\beta}.$$

This result does not depend on curvature because consumption is constant after the war.

13.3 Before-war Euler equation

Before the war, next period has no war with probability $1 - \pi$ and war with probability π . The Euler equation is

$$p_B = \beta \left[(1 - \pi)(z + p_B) + \pi \frac{u'(sz)}{u'(z)}(sz + p_A) \right].$$

13.4 Log utility

If $u(c) = \log c$, then $u'(sz)/u'(z) = 1/s$ and $sz + p_A = sz/(1 - \beta)$. Therefore

$$\begin{aligned} p_B &= \beta \left[(1 - \pi)(z + p_B) + \pi \frac{1}{s} \frac{sz}{1 - \beta} \right] \\ &= \beta \left[(1 - \pi)(z + p_B) + \pi \frac{z}{1 - \beta} \right]. \end{aligned}$$

Solving yields

Reusable formula

$$p_B = \frac{\beta}{1 - \beta}z.$$

This is the same as the no-war price.

The expected return before war is

$$\begin{aligned}\mathbb{E}[R_B] &= \frac{(1-\pi)(z+p_B) + \pi(sz+p_A)}{p_B} \\ &= \frac{(1-\pi)z/(1-\beta) + \pi sz/(1-\beta)}{\beta z/(1-\beta)} \\ &= \frac{1-\pi(1-s)}{\beta} < \frac{1}{\beta}\end{aligned}$$

when $\pi > 0$ and $s < 1$.

Economic intuition

With log utility, war risk lowers the expected cash payoff, but the war state is also a high-marginal-utility state. These two effects exactly offset in the price. The expected return is lower because the asset pays partly in a bad state and provides insurance value.

13.5 CRRA utility and risk-aversion comparison

For CRRA utility, $u'(c) = c^{-\sigma}$, so

$$\frac{u'(sz)}{u'(z)} = s^{-\sigma}.$$

Using $p_A = \beta sz/(1-\beta)$,

$$\begin{aligned}p_B &= \beta \left[(1-\pi)(z+p_B) + \pi s^{-\sigma} \frac{sz}{1-\beta} \right] \\ &= \beta \left[(1-\pi)(z+p_B) + \pi \frac{s^{1-\sigma}z}{1-\beta} \right].\end{aligned}$$

Thus

Reusable formula

$$\frac{p_B}{z} = \frac{\beta [(1-\pi) + \pi s^{1-\sigma}/(1-\beta)]}{1-\beta(1-\pi)}.$$

Compare this to the no-war price ratio $\beta/(1-\beta)$. The difference has the sign of $s^{1-\sigma} - 1$:

$$\frac{p_B}{z} - \frac{\beta}{1-\beta} = \frac{\beta\pi(s^{1-\sigma} - 1)}{(1-\beta)[1-\beta(1-\pi)]}.$$

Therefore:

- if $\sigma < 1$, war risk lowers the price;
- if $\sigma = 1$, war risk leaves the price unchanged;
- if $\sigma > 1$, war risk raises the price.

Economic intuition

When risk aversion is high, the asset's payoff in the war state is valued very highly because marginal utility is high. The insurance effect can dominate the lower physical dividend. When risk aversion is low, the cash-flow loss dominates.

14 Canonical model 7: bonds and liquidity value

14.1 Standard one-period risk-free bond

Add a one-period bond that costs $q(z)$ in state z and pays one unit of consumption next period. Let next bond holdings be b' . The budget with one stock is

$$c + p(z)y + q(z)b' \leq [z + p(z)]x + b.$$

The bond FOC is

$$u'(c)q(z) = \beta \mathbb{E}[V_b(y, b', z') \mid z].$$

Since $V_b(x, b, z) = u'(c)$ in the standard model,

$$q(z) = \beta \mathbb{E} \left[\frac{u'(c')}{u'(c)} \mid z \right] = \mathbb{E}[M(z, z') \mid z].$$

14.2 Bond-in-utility / liquidity value

A 2023-style problem gives preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t + \kappa b_t),$$

where b_t is the bond holding purchased in the previous period and carried into period t . The state is (x, b, z) and the choice is (y, b') :

$$V(x, b, z) = \max_{y, b'} \{u(c + \kappa b) + \beta \mathbb{E}[V(y, b', z') \mid z]\}$$

subject to

$$c + p(z)y + q(z)b' \leq [z + p(z)]x + b.$$

The stock FOC and bond FOC are

$$\begin{aligned} u'(c + \kappa b)p(z) &= \beta \mathbb{E}[V_x(y, b', z') \mid z], \\ u'(c + \kappa b)q(z) &= \beta \mathbb{E}[V_b(y, b', z') \mid z]. \end{aligned}$$

The envelope conditions are

$$\begin{aligned} V_x(x, b, z) &= u'(c + \kappa b)[z + p(z)], \\ V_b(x, b, z) &= u'(c + \kappa b)(1 + \kappa). \end{aligned}$$

In equilibrium, $x = y = 1$, $b = b' = 0$, and $c = z$. Therefore

$$\begin{aligned} u'(z)p(z) &= \beta \mathbb{E}[u'(z')(z' + p(z')) \mid z], \\ u'(z)q(z) &= \beta(1 + \kappa)\mathbb{E}[u'(z') \mid z]. \end{aligned}$$

14.3 Log closed forms

With $u(c) = \log c$,

Reusable formula

$$\begin{aligned} p(z) &= \frac{\beta}{1-\beta} z, \\ q(z) &= \beta(1+\kappa)z\mathbb{E}\left[\frac{1}{z'} \mid z\right], \\ R^f(z) &= \frac{1}{\beta(1+\kappa)z\mathbb{E}[1/z' \mid z]}, \\ \mathbb{E}[R^s \mid z] &= \frac{\mathbb{E}[z' \mid z]}{\beta z}. \end{aligned}$$

The expected stock return does not depend on κ , but R^f is decreasing in κ . Therefore the equity premium $\mathbb{E}[R^s \mid z] - R^f(z)$ is increasing in κ .

Economic intuition

The bond provides both a financial payoff and a liquidity/service payoff in utility. Households are willing to pay more for it. A higher bond price means a lower risk-free return. The stock price is unchanged in the log representative-agent equilibrium because stock holdings are fixed and stock does not enter utility directly.

15 Canonical model 8: time-varying net supply

15.1 Setup

A 2018-style problem has one asset with constant dividend $z > 0$ and total shares normalized to two. External investors exogenously demand $\hat{y}_t$, so domestic households must hold net supply

$$\bar{y}_t = 2 - \hat{y}_t.$$

Suppose

$$\bar{y}_t = \begin{cases} 2, & t \text{ even,} \\ 1, & t \text{ odd.} \end{cases}$$

The aggregate state must include parity because today's beginning holdings are last period's net supply while today's choice must equal today's net supply.

Let E denote an even period and O an odd period. Then

$$\begin{aligned} \bar{y}_-(E) &= 1, & \bar{y}(E) &= 2, \\ \bar{y}_-(O) &= 2, & \bar{y}(O) &= 1. \end{aligned}$$

In state $s \in \{E, O\}$, domestic consumption is

$$c(s) = z\bar{y}_-(s) + p(s)[\bar{y}_-(s) - \bar{y}(s)].$$

Thus

$$\begin{aligned} c(E) &= z - p_E, \\ c(O) &= 2z + p_O. \end{aligned}$$

15.2 Euler equations

The price equations are

$$\begin{aligned} u'(c(E))p_E &= \beta u'(c(O))(z + p_O), \\ u'(c(O))p_O &= \beta u'(c(E))(z + p_E). \end{aligned}$$

If u is linear, marginal utility is constant and

$$\begin{aligned} p_E &= \beta(z + p_O), \\ p_O &= \beta(z + p_E). \end{aligned}$$

The solution is

Reusable formula

$$p_E = p_O = \frac{\beta z}{1 - \beta}.$$

If u is strictly concave and the interior equilibrium has $c(O) > c(E)$, then $u'(c(E)) > u'(c(O))$. Let

$$a = \frac{u'(c(O))}{u'(c(E))} < 1.$$

Then

$$\begin{aligned}p_E &= \beta a(z + p_O), \\p_O &= \frac{\beta}{a}(z + p_E).\end{aligned}$$

Solving these two equations for given a gives $p_O > p_E$.

Economic intuition

The asset bought in an odd period pays in the next even period, when domestic households will have low consumption because they must absorb more shares. The payoff arrives in a high-marginal-utility state, so the odd-period price is high. The asset bought in an even period pays in a low-marginal-utility odd period, so its price is lower.

Qualifier trap

Do not set $x = y = 1$ in this problem. The whole point is that equilibrium household holdings change over time because external investors change net supply.

16 Canonical model 9: two countries, local stocks, and common bond

16.1 Setup template

There are two countries, $i = 1, 2$, each with one local tree. Country i residents can trade only stock i , but all residents can trade a common one-period bond. Let $z = (z_1, z_2)$ follow Markov transition Q . A country- i household enters with local stock holdings x_i and bond holdings b_i and chooses y_i, b'_i :

$$c_i + p_i(z)y_i + q(z)b'_i \leq [z_i + p_i(z)]x_i + b_i.$$

The Bellman equation is

$$V_i(x_i, b_i, z) = \max_{y_i, b'_i} \{u(c_i) + \beta \mathbb{E}[V_i(y_i, b'_i, z') \mid z]\}.$$

FOCs are

$$\begin{aligned} u'(c_i)p_i(z) &= \beta \mathbb{E}[V_{i,x}(y_i, b'_i, z') \mid z], \\ u'(c_i)q(z) &= \beta \mathbb{E}[V_{i,b}(y_i, b'_i, z') \mid z]. \end{aligned}$$

Envelopes are

$$\begin{aligned} V_{i,x}(x_i, b_i, z) &= u'(c_i)[z_i + p_i(z)], \\ V_{i,b}(x_i, b_i, z) &= u'(c_i). \end{aligned}$$

Thus

$$\begin{aligned} u'(c_i)p_i(z) &= \beta \mathbb{E}[u'(c'_i)(z'_i + p_i(z')) \mid z], \\ u'(c_i)q(z) &= \beta \mathbb{E}[u'(c'_i) \mid z]. \end{aligned}$$

16.2 Equilibrium objects

A stationary recursive competitive equilibrium consists of value functions V_i , policy functions g_i^x, g_i^b, c_i , price functions $p_i(z)$ and $q(z)$, and a bond distribution or country bond positions satisfying household optimization, stock market clearing $y_i = 1$, bond market clearing $b'_1 + b'_2 = 0$, and feasibility $c_1 + c_2 = z_1 + z_2$.

16.3 Alternating-dividend closed-form benchmark

Suppose tree 1 pays 1 in even periods and 0 in odd periods; tree 2 pays 0 in even periods and 1 in odd periods. If the bond supports constant consumption for each country, marginal utility is constant. The bond price is

$$q = \beta.$$

For tree 1, let p_E be the ex-dividend price in an even period and p_O in an odd period. Since the next dividend after an even period is 0 and the next dividend after an odd period is 1,

$$\begin{aligned} p_E &= \beta p_O, \\ p_O &= \beta(1 + p_E). \end{aligned}$$

Thus

Reusable formula

$$p_O = \frac{\beta}{1 - \beta^2}, \quad p_E = \frac{\beta^2}{1 - \beta^2}.$$

Tree 2 has the opposite price pattern.

Economic intuition

The bond smooths consumption across countries, making marginal utility constant in the closed-form case. Stock prices then reduce to discounted dividend timing. This is why the concave-utility-with-bond prices can coincide with linear-utility prices.

17 Reusable derivation templates

17.1 Template A: Bellman to Euler

- A1.** Budget: $c + p(s)y \leq R_x(s)x + T(s)$, where $R_x(s)$ is the current value of beginning holdings.
A2. Bellman: $V(x, s) = \max_y \{u(c) + \beta \mathbb{E}[V(y, s') \mid s]\}$.
A3. FOC: $u'(c)p(s) = \beta \mathbb{E}[V_x(y, s') \mid s]$.
A4. Envelope: $V_x(x, s) = u'(c)R_x(s)$.
A5. Euler: $u'(c)p(s) = \beta \mathbb{E}[u'(c')R_x(s') \mid s]$.
A6. Equilibrium: substitute $x = y = \text{fixed supply}$ and $c = \text{aggregate dividend net of resource effects}$.

17.2 Template B: log price guess

If $u(c) = \log c$, write the Euler equation as

$$\frac{p(s)}{c(s)} = \beta \mathbb{E} \left[\frac{\text{payoff}(s')}{c(s')} \mid s \right].$$

Then define $f(s) = p(s)/c(s)$ or $f(s) = u'(c(s))p(s)$ and solve the linear equation

$$f(s) = h(s) + \beta \mathbb{E}[f(s') \mid s].$$

If $h(s)$ is constant, $f = h/(1 - \beta)$.

17.3 Template C: two-state spread

For states H, L with persistence π and $\rho = 2\pi - 1$, if

$$p_s = \beta \mathbb{E}[d_{s'} + p_{s'} \mid s]$$

and $d_H - d_L = \Delta d$, then

$$p_H - p_L = \frac{\beta \rho}{1 - \beta \rho} \Delta d.$$

This template also works for tax-state spreads after replacing Δd with the relevant after-tax payoff difference.

17.4 Template D: CRRA growth

When $d_{t+1} = \gamma d_t$, $c_t = d_t$, and $u'(c) = c^{-\sigma}$:

1. Write $p_t = \beta \gamma^{-\sigma} (d_{t+1} + p_{t+1})$.
2. Guess $p_t = \nu d_t$.
3. Solve $\nu = \beta \gamma^{1-\sigma} (1 + \nu)$.
4. Check $\beta \gamma^{1-\sigma} < 1$.
5. Return: $R = \gamma^\sigma / \beta$.

18 Closed-form catalog with interpretation

Case	Formula	Intuition / use
Generic SDF	$p_i(z) = \mathbb{E}[M(z, z')(d'_i + p_i(z')) \mid z]$	Every model is this equation with a modified payoff or modified M .
Risk-free bond	$q(z) = \mathbb{E}[M(z, z') \mid z], R^f = 1/q$	A risk-free payoff is priced by the expected SDF.
Risk premium	$\mathbb{E}[R_i] - R^f = -R^f \text{Cov}(M, R_i)$	Good hedge means positive covariance with M and low expected return.
Log one-tree	$p(z) = \frac{\beta}{1-\beta} z$	Current high consumption lowers current marginal utility and raises the price proportionally.
Linear utility	$p_i(z) = \sum_{s \geq 1} \beta^s \mathbb{E}[d_{i,t+s} \mid z_t = z]$	No risk adjustment; only expected discounted dividends matter.
Two negatively correlated trees	$p_1(z) + p_2(z) = \frac{\beta}{1-\beta}$	Aggregate consumption is constant, so total equity value is constant.
Binary persistence spread	$\Delta p = \frac{\beta(2\pi-1)}{1-\beta(2\pi-1)} \Delta d$	More persistence makes current dividends more informative about future dividends.
Dividend-tax spread	$p_L - p_H = \frac{\beta(2\theta-1)\bar{\tau}\bar{z}}{1-\beta(2\theta-1)}$	A current low tax raises price only if it predicts low future taxes.
Tax on asset value	$p(z) = \frac{\beta(1-\tau)}{1-\beta(1-\tau)} z$	Private payoff is scaled by $1 - \tau$, so price falls.
CRRA growth	$p_t = \frac{\beta\gamma^{1-\sigma}}{1-\beta\gamma^{1-\sigma}} d_t$	Growth raises future cash flows but lowers future marginal utility.
Growth return	$R = \gamma^\sigma / \beta$	Higher growth raises required return under CRRA.
War after price	$p_A = \frac{\beta}{1-\beta} sz$	After war, consumption is constant.
War before CRRA	$p_B/z = \frac{\beta[(1-\pi)+\pi s^{1-\sigma}/(1-\beta)]}{1-\beta(1-\pi)}$	Price can rise or fall with disaster risk depending on σ .
Bond-in-utility log	$q(z) = \beta(1 + \kappa)z\mathbb{E}[1/z' \mid z]$	Liquidity value raises bond price and lowers risk-free rate.

19 Exam variations and how to mutate the canonical model

Variation	What changes in the setup	What stays invariant
Multiple trees	x, y, p, d become vectors; aggregate consumption is sum of dividends.	FOC-envelope-Euler for each asset.
Negatively correlated dividends	Individual dividends move, but aggregate consumption may be constant.	Use aggregate $u'(c)$ in SDF.
Dividend taxes	Payoff becomes $(1 - \tau')d' + p'$. Rebate enters budget but cancels in aggregate consumption.	Current price still equals discounted next payoff.
Tax on beginning asset value	Envelope is multiplied by $1 - \tau$.	FOC still has current purchase price on the left.
Deterministic growth	Problem is nonstationary in levels. Normalize by dividend.	Euler equation still prices next dividend plus resale price.
War risk	Add before-war and after-war states; transition from before to after is absorbing.	Same SDF logic; bad-state payoff can be valuable.
Risk-free bond	Add qb' to budget and bond payoff b to resources.	Bond Euler is the asset Euler for a safe payoff.
Bond in utility	Current bond holding enters utility directly, so V_b includes $1 + \kappa$.	Stock Euler unchanged unless stock enters utility or affects consumption.
Time-varying supply	Aggregate state includes net supply/parity; market clearing is $y = \bar{y}(s)$, not one.	FOC-envelope-Euler still applies to individual problem.
Two countries and market segmentation	Households have local stock and common bond; bond positions become state/distribution variables.	Each country's Euler equation uses its own marginal utility.

20 Comparative statics playbook

20.1 Persistence

Persistence affects prices when the current state predicts future payoffs. In two-state problems, reduce the price comparison to a spread equation. If $\rho = 2\pi - 1$ and the high state has a higher future payoff, then price spreads are increasing in ρ :

$$\Delta p = \frac{\beta\rho}{1 - \beta\rho} \Delta \text{payoff}.$$

If the current state predicts a lower future payoff, the sign flips.

20.2 Growth

Higher growth has two forces:

- cash-flow effect: future dividends rise;
- discounting effect: future consumption rises, lowering future marginal utility.

CRRA curvature decides the sign. Use $\beta\gamma^{1-\sigma}$.

20.3 Risk and disasters

Risk is not automatically bad for price. Ask whether the asset pays in high-marginal-utility states. War lowers physical dividends, but it also raises marginal utility. With CRRA, the comparison with the no-war price is governed by $s^{1-\sigma} - 1$.

20.4 Taxes

Separate taxes that change private asset payoffs from taxes that change aggregate feasibility. With fixed tree supply and lump-sum rebates, a tax can lower prices without changing the allocation.

20.5 Liquidity value

If bonds enter utility, the bond has a service payoff in addition to its financial payoff. Higher liquidity value raises bond prices, lowers risk-free rates, and can raise measured equity premia even when stock prices and expected stock returns are unchanged.

21 Common traps

- Trap 1. Wrong timing.** The price p_t is ex-dividend. Today's buyer receives tomorrow's dividend, not today's dividend.
- Trap 2. Equilibrium too early.** Do not set $x = y = 1$ before deriving the FOC and envelope.
- Trap 3. Missing aggregate consumption.** In multi-tree economies, $u'(c)$ depends on total consumption, not on one tree's dividend.
- Trap 4. Ignoring rebates.** Dividend taxes with lump-sum rebates often leave $c = z$ in equilibrium even though private payoffs are after tax.
- Trap 5. Using current tax instead of next tax.** The payoff in the Euler equation uses τ' .
- Trap 6. Confusing price and return.** War risk can leave the log price unchanged but lower the expected return.
- Trap 7. Claiming higher growth always raises price.** Under CRRA, the price-dividend ratio falls with growth when $\sigma > 1$.
- Trap 8. Forgetting bond zero net supply.** In equilibrium, $b = 0$ for the representative household or aggregate bond holdings sum to zero.
- Trap 9. Missing liquidity in V_b .** If utility is $u(c + \kappa b)$, then $V_b = u'(c + \kappa b)(1 + \kappa)$.
- Trap 10. Calling every tax inefficient.** Inefficiency is about allocations and wedges, not about prices alone.

22 Exercise bank

22.1 Mechanical drills

- M1. Skill: write the Bellman equation.** Write the recursive household problem for a k -tree Lucas economy with Markov state z and price vector $p(z)$.
- M2. Skill: FOC-envelope-Euler.** Starting from the k -tree Bellman equation, derive the FOC, envelope condition, and equilibrium Euler equation for asset i .
- M3. Skill: SDF and bond price.** Given $p(z) = \mathbb{E}[M(z, z')(d' + p(z')) \mid z]$, derive $1 = \mathbb{E}[MR]$ and $q(z) = \mathbb{E}[M \mid z]$.
- M4. Skill: tax timing.** A dividend tax τ_t is observed before trade and rebated lump-sum. Write the household budget and equilibrium Euler equation.
- M5. Skill: bond-in-utility setup.** Utility is $u(c + \kappa b)$ where b is the bond holding carried into the period. Write the Bellman equation and derive V_b .
- M6. Skill: RCE objects.** List the objects in a stationary recursive competitive equilibrium for a stock-and-bond Lucas economy.

22.2 Closed-form and proof drills

- C1. Skill: log one-tree.** Prove that $p(z) = \beta z / (1 - \beta)$ for a one-tree log economy with arbitrary Markov dividends.
- C2. Skill: two-tree price sum.** In the negatively correlated two-tree economy $d_1 = z$, $d_2 = 1 - z$, show that $p_1(z) + p_2(z) = \beta / (1 - \beta)$.
- C3. Skill: persistence spread.** For binary $z \in \{H, L\}$ with persistence π , derive $p_1(H) - p_1(L)$.
- C4. Skill: tax-state spread.** In the PS2 two-tax-state log case, derive $p_L - p_H$ and sign it.
- C5. Skill: CRRA growth.** Derive $p_t = \nu d_t$, $\nu = \beta \gamma^{1-\sigma} / [1 - \beta \gamma^{1-\sigma}]$, and $R = \gamma^\sigma / \beta$.
- C6. Skill: war risk.** Derive the before-war CRRA price and compare it with the no-war price.
- C7. Skill: bond liquidity.** Under log utility and bond-in-utility, derive $p(z)$, $q(z)$, $R^f(z)$, and $\mathbb{E}[R^s \mid z]$.
- C8. Skill: iid two-tree.** Tree 1 has iid dividend x , tree 2 has constant dividend 1, and utility is log. Derive $p_1(x)$ and $p_2(x)$.

22.3 Past-qual-style applications

- P1. 2016-style: two trees.** Formulate the recursive problem and RCE for two Markov trees. Then solve the iid-log one-constant-tree special case.
- P2. 2017-style: persistence.** In the negatively correlated binary-dividend economy, explain formally how higher persistence affects price volatility.
- P3. 2018-style: time-varying net supply.** Set up the odd/even state problem and compare odd/even prices under linear and strictly concave utility.
- P4. 2019-style: two countries and bond.** Write the recursive problems for both countries, define stationary RCE, and solve the alternating-dividend constant-consumption benchmark.
- P5. 2021-style: tax on income plus asset value.** Show prices decrease with τ and evaluate the claim that the tax makes the allocation inefficient.
- P6. 2023-style: bond-in-utility.** Derive log closed forms and explain whether the equity premium is increasing in κ .

22.4 Trap and economist-claim questions

- T1.** Economist A says: “The asset price must be higher in the zero-tax state.” Evaluate in the PS2 tax model.
- T2.** Economist B says: “Higher dividend growth always raises the equilibrium asset price.” Evaluate under CRRA.
- T3.** Economist C says: “War risk lowers the asset price because dividends may fall.” Evaluate under CRRA.
- T4.** Economist D says: “Any distortionary tax on asset payoffs makes the competitive allocation inefficient.” Evaluate in the fixed-tree rebate economy.
- T5.** Economist E says: “If today’s dividend is high, the asset price must be high.” Evaluate using income and information effects.

23 Answer sketches

23.1 Mechanical drills

M1. State (x, z) , choice y , budget $c + p(z) \cdot y \leq [z + p(z)] \cdot x$, Bellman

$$V(x, z) = \max_y \{u(z \cdot x + p(z) \cdot (x - y)) + \beta \mathbb{E}[V(y, z') \mid z]\}.$$

Common mistake: using only one asset holding when there are k trees.

M2. FOC: $u'(c)p_i(z) = \beta \mathbb{E}[V_{x_i}(y, z') \mid z]$. Envelope: $V_{x_i} = u'(c)[z_i + p_i(z)]$. Equilibrium: $c = \sum_j z_j$, $x = y = \mathbf{1}$, so

$$u'(c(z))p_i(z) = \beta \mathbb{E}[u'(c(z'))(z'_i + p_i(z')) \mid z].$$

Common mistake: envelope missing z_i .

M3. Return $R' = (d' + p')/p$. Divide the price equation by p to get $1 = \mathbb{E}[MR']$. For a risk-free payoff of one, $q = \mathbb{E}[M]$ and $R^f = 1/q$. Common mistake: using $1/\mathbb{E}[R]$ as the bond price.

M4. Budget: $c + p(z, \tau)y \leq [p(z, \tau) + (1 - \tau)z]x + T(z, \tau)$. Equilibrium $T = \tau z$, $c = z$. Euler:

$$u'(z)p(z, \tau) = \beta \mathbb{E}[u'(z')((1 - \tau')z' + p(z', \tau')) \mid z, \tau].$$

Common mistake: using $(1 - \tau)z$ in the next payoff.

M5. Bellman:

$$V(x, b, z) = \max_{y, b'} \{u(c + \kappa b) + \beta \mathbb{E}[V(y, b', z') \mid z]\}$$

subject to $c + py + qb' \leq (z + p)x + b$. Since b raises resources and directly enters utility, $V_b = u'(c + \kappa b)(1 + \kappa)$. Common mistake: writing $V_b = u'$.

M6. Objects: V , policies c, g^x, g^b , prices p, q , transition law for z , and initial conditions. Conditions: household optimization, stock clearing $g^x = 1$, bond clearing $g^b = 0$ in representative-agent equilibrium, feasibility $c = z$, and price consistency. Common mistake: omitting bond market clearing.

23.2 Closed-form and proof drills

C1. With log utility, $p(z)/z = \beta \mathbb{E}[1 + p(z')/z' \mid z]$. Let $f = p/z$. Then $f = \beta + \beta \mathbb{E}f$. The unique bounded solution is $f = \beta/(1 - \beta)$.

C2. Since $c = 1$, marginal utility is constant. Add the equations $p_1 = \beta \mathbb{E}[z' + p'_1]$ and $p_2 = \beta \mathbb{E}[1 - z' + p'_2]$. The sum S solves $S = \beta(1 + S)$, so $S = \beta/(1 - \beta)$.

C3. Let $\rho = 2\pi - 1$ and $\Delta d = H - L$. Subtract low-state from high-state price equations: $\Delta p = \beta \rho(\Delta d + \Delta p)$. Hence $\Delta p = \beta \rho \Delta d / (1 - \beta \rho)$. Persistence raises the spread.

C4. The tax equations are

$$p_L = \beta[\theta(\bar{z} + p_L) + (1 - \theta)((1 - \bar{\tau})\bar{z} + p_H)]$$

and

$$p_H = \beta[(1 - \theta)(\bar{z} + p_L) + \theta((1 - \bar{\tau})\bar{z} + p_H)].$$

Subtract to obtain $p_L - p_H = \beta(2\theta - 1)(\bar{\tau}\bar{z} + p_L - p_H)$. The sign is positive iff $\theta > 1/2$.

C5. Guess $p_t = \nu d_t$. The Euler gives $\nu = \beta \gamma^{1-\sigma}(1 + \nu)$, so $\nu = \beta \gamma^{1-\sigma} / [1 - \beta \gamma^{1-\sigma}]$. Return is $R = \gamma(1 + \nu)/\nu = \gamma^\sigma / \beta$.

C6. After war, $p_A = \beta sz/(1 - \beta)$. Before war,

$$p_B = \beta[(1 - \pi)(z + p_B) + \pi s^{-\sigma}(sz + p_A)].$$

Solve to get

$$p_B/z = \frac{\beta[(1 - \pi) + \pi s^{1-\sigma}/(1 - \beta)]}{1 - \beta(1 - \pi)}.$$

Compare to $\beta/(1 - \beta)$; sign is $s^{1-\sigma} - 1$.

C7. Log stock: $p(z) = \beta z/(1 - \beta)$. Bond-in-utility: $q(z) = \beta(1 + \kappa)z\mathbb{E}[1/z' | z]$. Thus $R^f = 1/q$ and $\mathbb{E}[R^s | z] = \mathbb{E}[z' | z]/(\beta z)$. Equity premium rises in κ .

C8. Guess $p_i(x) = a_i(x + 1)$. Then $a_1 = \beta\mathbb{E}[x'/(x' + 1)] + \beta a_1$ and $a_2 = \beta\mathbb{E}[1/(x' + 1)] + \beta a_2$. Solve as in the catalog. Current x raises both prices through the current marginal-utility denominator.

23.3 Past-qual-style applications

- P1.** Use the generic $k = 2$ Bellman. RCE: price functions p_1, p_2 , value function, policies, transition law, market clearing $y_1 = y_2 = 1$, feasibility $c = x_1 + x_2$ in dividend units. The iid-log special case uses $p_i(x) = a_i(x + 1)$.
- P2.** Price volatility is $|p_1(H) - p_1(L)| = \frac{\beta(2\pi-1)}{1-\beta(2\pi-1)}(2\bar{z} - 1)$, increasing in π for $\pi > 1/2$. Intuition: current high dividend predicts future high dividend more strongly.
- P3.** State is parity. Linear utility gives $p_E = p_O = \beta z/(1 - \beta)$. Strict concavity gives $c(E) < c(O)$, so the odd price is higher because the odd purchase pays in the low-consumption even state.
- P4.** For each country, write stock and bond FOCs. In the alternating benchmark with constant marginal utility, $q = \beta$ and tree 1 prices satisfy $p_E = \beta p_O$, $p_O = \beta(1 + p_E)$. Thus $p_O = \beta/(1 - \beta^2)$ and $p_E = \beta^2/(1 - \beta^2)$.
- P5.** Euler: $u'(z)p(z) = \beta(1 - \tau)\mathbb{E}[u'(z')(z' + p(z'))]$. Log price is $p(z) = \frac{\beta(1-\tau)}{1-\beta(1-\tau)}z$, decreasing in τ . Allocation is still $c = z$ with fixed supply and rebates, so price distortion alone does not prove inefficiency.
- P6.** $p(z)$ is independent of κ under log, $q(z)$ increases in κ , and R^f decreases. Expected stock return is unchanged. Hence the equity premium increases in κ .

23.4 Trap and economist-claim questions

- T1.** Economist A is wrong in general. $p_L - p_H$ has the sign of $2\theta - 1$. Zero tax today raises price only if zero tax is persistent.
- T2.** The claim is false. For fixed current dividend, $\partial(p_t/d_t)/\partial\gamma$ has sign $1 - \sigma$. Higher growth raises future dividends but lowers future marginal utility.
- T3.** The claim is incomplete. War risk lowers price if $\sigma < 1$, leaves it unchanged if $\sigma = 1$, and raises it if $\sigma > 1$. Bad-state payoffs have insurance value.
- T4.** Not necessarily. In the fixed-tree representative-agent economy with lump-sum rebates, the allocation can remain $c = z$. Taxes lower private prices but need not create an allocation wedge.
- T5.** Not always. A high current dividend lowers current marginal utility and creates an income effect that tends to raise prices. But if high current dividends predict low future dividends, the information effect can lower prices. The sign is model-dependent.

24 Past-qual practice map

Year / question	Asset-pricing content	What to practice from this packet	Priority
2016 Q3	Two types of trees; recursive problem; RCE; log iid tree 1 and constant tree 2; show prices higher in high-dividend state.	Generic two-tree Bellman; iid-log two-tree closed form; income-effect intuition.	High
2017 Q1	Two negatively correlated trees; constant aggregate dividends; price sum; binary persistence comparative statics.	Constant aggregate consumption model; binary spread formula.	High
2018 Q1	Time-varying net supply due to external investors; odd/even prices; linear vs concave utility.	Time-varying supply setup; parity state; marginal-utility price ranking.	High
2019 Q1	Two countries; segmented stock markets; common bond; FOC/envelope/Euler; alternating dividends closed form.	Stock-and-bond setup; country-specific Euler equations; bond clearing.	Medium-high
2020 Q2	Search with asset market and bond.	Mostly Module 4; use only bond budget discipline.	Low for Module 3
2021 Q2	Lucas tree with tax on dividends plus asset value; price decreasing in tax; log closed form; efficiency claim.	Tax-on-asset-value model; monotonicity; price vs allocation distinction.	High
2022	No direct Lucas-tree asset-pricing problem.	Dynamic risk intuition only.	Low
2023 Q2	Risky stock plus risk-free bond; bond in utility; log closed forms; equity premium and κ .	Bond-in-utility model; risk-free rate; equity premium.	High
2024	No direct asset-pricing problem.	None.	Low
PS2 2026	Markov dividend/tax recursive problem; log tax persistence; CRRA growth; war risk.	Treat as current benchmark problem set.	Very high

25 Final study checklist

Before attempting old asset-pricing qual problems, make sure you can do the following without notes:

- ✓1. Write the ex-dividend budget constraint for one tree and for k trees.
- ✓2. Derive FOC, envelope, and Euler in under three minutes.
- ✓3. State the SDF and explain why bad-state payoffs are valuable.
- ✓4. Solve the log one-tree price $p(z) = \beta z / (1 - \beta)$.
- ✓5. Solve a two-state linear price system by subtracting high and low equations.
- ✓6. Explain why tax-state price rankings depend on persistence.
- ✓7. Derive the CRRA growth price-dividend ratio and return.
- ✓8. Explain why higher growth may lower the price-dividend ratio.
- ✓9. Derive before-war and after-war prices and compare by σ .
- ✓10. Add a bond to the budget and derive $q(z) = \mathbb{E}[M]$.
- ✓11. Add bond-in-utility and remember $V_b = u'(c + \kappa b)(1 + \kappa)$.
- ✓12. Track changing net supply with an aggregate state rather than forcing $x = y = 1$.
- ✓13. Distinguish price effects from allocation efficiency.

26 Quality audit

- Source inventory is explicit and separates current problem sets, professor notes, and past quals.
- The packet follows the required asset-pricing workflow: recursive household problem, budget timing, FOC, envelope, Euler, market clearing, SDF, closed forms, expected returns, and risk-free rates.
- Current PS2 emphasis appears in Markov taxes, CRRA growth, expected return comparisons, and war risk.
- Past-qual patterns from 2016, 2017, 2018, 2019, 2021, and 2023 are mapped and converted into reusable templates.
- Closed forms are derived rather than merely stated.
- Exercises are labeled by skill and answer sketches include setup, key equation, result, intuition, and common mistakes.
- Remaining limitation: exact page numbers for local TeX past exams are not included because the uploaded past-exam sources were TeX files rather than paginated PDFs.